# Palaeomagnetic Directions of the Deccan Traps of Rajahmundry, Andhra Pradesh, India

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*Summary*

The palaeomagnetic directions of the Deccan traps of Rajahmundry in Andhra Pradesh, India which are considered to be an outlier of the main Deccan traps of western and central India supports the evidence that the Indian sub-continent has drifted northwards across the equator and rotated anticlockwise since the formation of the traps, as has been postulated by earlier workers. However these rocks are found to be normally magnetized (dip up 45° and declination 305° E) while a rock series, the Linga rocks, in the main Deccan trap area, considered to be contemporaneous with the Rajahmundry traps are reversely magnetized (dip down 48°, declination 164° E). Three possibilities are suggested by these conflicting results: (1) The Linga and Rajahmundry rocks are not geologically contemporaneous with one another. (2) Though belonging to the same age group they are separated by one or two million years and the Earth's field had been reversed in between their eruption and (3) both rocks erupted simultaneously. One, however, had self reversed and this self reversing mechanism had later been obscured. Of these possibilities, the second appears the more probable from both geological and palaeomagnetic evidence.

## 1. Introduction

Considerable work on the palaeomagnetic directions of the Indian rocks had been undertaken earlier by Professor Blackett of the Physics Department of Imperial College, London, and by the Tata Institute of Fundamental Research, Bombay. Most of the work has so far been confined to the Deccan traps of western India (Irving 1956, Clegg & others 1956, Deutsch & others 1958) though recently attempts have been made by Athavale & others (1963) to extend the work to the rocks of the other parts of the sub-continent, as well as to the older rocks. The present paper deals with the result of the palaeomagnetic directions of the Deccan traps of Rajahmundry (near the east coast of India—longitude 81° 47′ and latitude 17° 03′) which are considered to be outlying areas of the original Deccan traps.

## 2. Geology

The traps of Rajahmundry area are believed to be parts of the main Deccan trap area of western and central India, the original extent of which was far greater

than the extent of the present exposures (Krishnan 1956). The Rajahmundry traps consisting of two or possibly three flows appear on either bank of river Godavari, with a length of about 35 miles in ENE–WSW direction and a thickness of 100 to 200 feet. Exposures are found near Kateru (4 miles from Rajahmundry) resting on the Archean rocks and near Pangidi and Duddukuru on the other side of the river, resting often on upper Gondwanas. The traps are overlain by Rajahmundry sandstones and generally show a shallow dip towards south or south-east. A recent magnetic survey carried out by the research workers of the Geophysics Department of Andhra University indicates the possibility of the trap extending below the sandstones over a larger area than originally thought.

The Deccan traps are believed to be the result of volcanic activity at the close of the Mesozoic era, marked by the outpouring of enormous lava flows through narrow fissures in the Earth's crust. The traps can be divided into three broad groups—Upper, Middle and Lower—with Inter-trappeans or Lameta beds at their bases. It is believed that there was a considerable time lag between different eruptions, the upper traps being probably middle or upper Eocene. The Rajahmundry traps are believed to be of early Eocene age and thus represent the lower series of the traps (Narayanarao & Sripadarao 1936). Venkayya (1949) concludes from a similarity in chemical composition with the traps of the Linga area (in the Chindwara district, Madhya Pradesh, mean latitude 21° 58′ N, mean longitude 78° 55′ E) that the traps of Rajahmundry and Linga areas are contemporaneous, representing the lower traps.

## 3. Collection of samples and measurement

Thirty-five oriented samples of the traps were collected from the available fresh exposures in the quarries near Kateru during September 1962 and were shipped to London for measurement. Each sample was cut into cylindrical specimens of $\frac{7}{8}$ in. diameter and $\frac{3}{4}$ in. length, an average of two to three specimens being obtained from each sample. The magnetic directions of the specimens were measured with an astatic magnetometer. Six of the specimens have been tested for their magnetic stability by heating them in zero field under an astatic magnetometer and observing their magnetic moment in three perpendicular directions as has been described by Wilson (1962). For a typical specimen the variation of saturation moment (in 6 000 Oe) with temperature was measured employing the principle of force on the specimen in a non-uniform magnetic field.

## 4. Results

In Figure 1 are plotted the mean palaeomagnetic direction for each sample, on a Schmidt's projection. Only the consistent high temperature magnetic directions for the six thermally demagnetized samples are plotted in this figure. As can be seen from this figure the magnetic directions of the rocks are reasonably consistent and give a mean direction of 305° E and 45° up. The magnetic intensity of the rocks varied from 0·5 to $3{\cdot}5 \times 10^{-3}$ e.m.u./g. The declination of 305° E and the upward inclination of the rocks which differ very much from the present declination and dip of the Earth's field near Rajahmundry can be explained by the assumption that the Indian sub-continent has moved northwards across the equator relative to the Earth's geographical axis and rotated in an anticlockwise direction since these rocks erupted and were laid down. This finding is in general agreement with the

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results of earlier workers on the Deccan traps of western India (Clegg & others 1956, 1958). On the assumption that the mean magnetic direction of the rocks corresponds to that of the central magnetic dipole coaxial with the Earth's rotational axis, one could calculate the ancient latitude ($\lambda$) of the site at the time of eruption of the rocks, using the formula $\tan \lambda = \frac{1}{2} . \tan I$ where $I$ is the mean dip of the rocks. The ancient latitude obtained in this case is 27° south and is in agreement with the ancient latitude calculated for the Linga traps (29° S) and thus supports the geological evidence of the contemporaneity of the Rajahmundry and Linga traps.

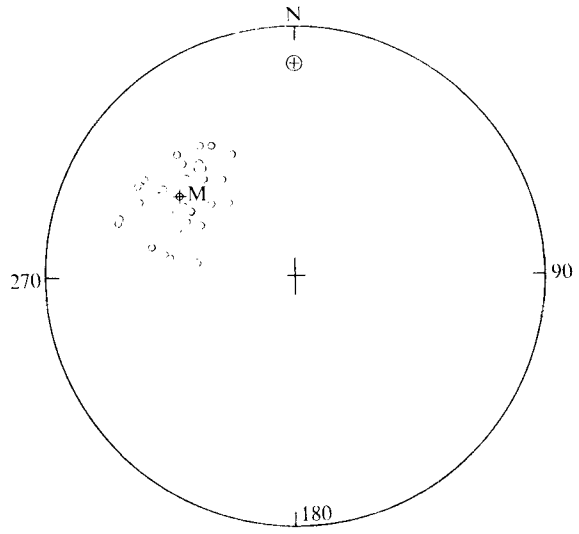



FIG. 1.—Magnetic directions of Rajahmundry Deccan traps—Schmidt's equal area projection (All dips up).

⊕ Mean direction
⊕ Present dipole

The results of the earlier workers on the Deccan traps of Western India while giving support to the theory that the Indian sub-continent has moved northwards and rotated in an anticlockwise direction, showed that the directions of magnetization of certain sequences of traps are "normal" (dip up and declination around 300° E) while certain other sequences are "reversed" (dip down and declination around 130° E). Thus the present rocks can be classified as "normal" on the assumption that the Earth's field was in the same direction as it is now, at the time of eruption of these rocks, and that the present upward dip and different declination are simply brought about by the indicated drift of India, while the Linga rocks showing almost antiparallel magnetic directions are to be classified as "reversed" rocks.

In Figure 2 the variations of the natural moment of a typical specimen in three perpendicular directions are plotted. They are obtained on heating the rocks in zero field under the astatic magnetometer. The variation of the direction of the magnetic vector with temperature in the horizontal and vertical planes is shown in Figure 3. It will be observed from Fig. 3 that the direction of the magnetic vector rotates up to about 270 °C when it steadies itself and more or less continues to

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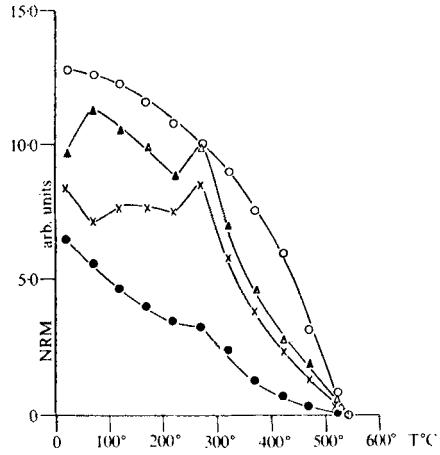



FIG. 2.—Temperature variation of NRM in zero field in three perpendicular directions and the TRM acquired in 0·5 Oe.

● X component
× Y component } NRM
△ Z component

○ TRM in 0·5 Oe

$$\frac{J_N}{J_{TRM}} = 0{\cdot}22$$

drop in a straight line to zero at the curie point. The rotation of the magnetic vector below 270 °C can probably be attributed to secondary magnetization components (due to viscous or other magnetization) while a constant direction is given by the magnetic vectors above this temperature. The same effect could be observed in the variation of natural magnetic moment with temperature (Figure 2). Both the *Y* and *Z* components in the above figure show an erratic variation in magnetic moment with increasing temperature up to 270 °C and the *X* component shows a discontinuity in gradient around 270 °C; above this they all gradually fall to zero at the curie point (around 550 °C). In all the six specimens examined (for magnetic

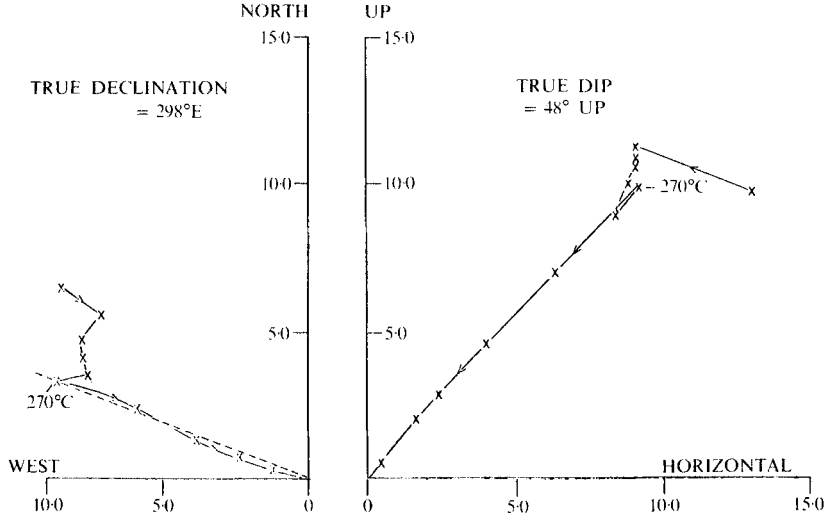



FIG. 3.—Temperature variation of the NRM vector along horizontal and vertical planes.

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stability) a definite change in gradient around 220–270 °C existed. The room-temperature magnetic directions and those above 270 °C do not however differ significantly from one to another, which suggests that the secondary magnetization effects are small and so the rocks can be considered reasonably stable. One sample, whose magnetic direction changed persistently throughout heating to the curie point, was considered unstable and was rejected.

The thermoremanence of these rocks in 0·5 Oe (also plotted in Figure 2) is always greater than their natural moment and at times is greater by a factor of ten. It increases gradually from the curie point to room temperature without any discontinuity in the 270–220 °C region as is present in the natural moment during first heating.

Figure 4 shows the variation of the magnetic moment (in 6 000 Oe) with the temperature of a specimen (for a specimen heated in air). The curve obtained during the first heating shows a clear discontinuity at 320 °C above which the curve has a different shape and drops off to zero at 560 °C. Thus in high fields it appears that these rocks show two curie points (320 and 560 °C). On heating to and cooling from 560 °C, the cooling curve shows only the higher curie point at 560 °C. The room temperature magnetic moment of the heated sample (in 6 000 Oe) is about twice that of the fresh sample while for the same sample the ratio of the TRM in 0·5 Oe to the natural moment is around 10.

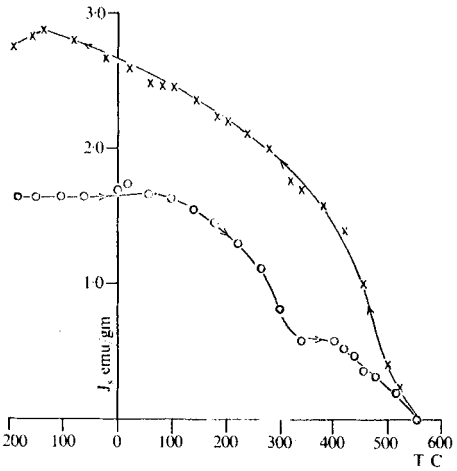



FIG. 4.—Variation of saturation moment (in 6 000 Oe) with temperature.

○ Heating × Cooling

## 5. Discussion and conclusions

The mean magnetic direction of the rocks from Rajahmundry is 305° E and 45° up. This direction is in agreement with the drift of the Indian peninsula as has been proposed by earlier workers, from the palaeomagnetic directions of the Deccan traps of western India. However while the Rajahmundry rocks are normally magnetized, the rocks from the Linga area (with magnetic directions around 164° E and 48° down), presumably contemporaneous with the present rocks, are seen to be reversely magnetized. From the palaeomagnetic results which they obtained, Deutsch & others proposed that rocks belonging to the upper division of the Deccan traps were normally magnetized and to the lower reversely magnetized. They indicated the possible application of the palaeomagnetic results to dating those rocks

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which could not be stratigraphically mapped because of lack of fossil records. The normal and the reverse traps, according to Deutsch & others, represent the reversal of the Earth's field from time to time during the eruption of the traps. The present normally-magnetized Rajahmundry rocks should represent, according to the above time scale, the upper normal series though the geological evidence of these rocks show that they belong to the lower traps and are of the same age as the Linga traps. It appears therefore that the present rocks do not fit into the above mentioned classification of Deutsch & others.

If we accept that the Earth's field has periodically reversed during the eruption of the Deccan traps, the present rocks might have acquired their magnetization when the Earth's field was normal, while the Linga traps might have acquired their magnetization when the Earth's field was reversed. The contemporaneity of the Linga traps with these rocks need not be taken literally. Since the accepted period of reversal of Earth's field may vary from 1 My to 20 or more My, the rocks, though geologically in the same division, might acquire opposite directions of magnetization if the Earth's field had reversed between the periods of their respective eruptions. An alternative, if we accept the contemporaneity of Rajahmundry traps with Linga traps, is that one of them could have been self-reversed owing to an inherent mechanism in the rocks. In the present rocks there is no evidence of self-reversal, at least as far as is shown in the laboratory tests carried out so far. The change in the gradient of the variation of the natural moment of these rocks with temperature at 270 °C (Figure 2) is assumed to be due to some form of secondary magnetization.

On the other hand the first heating curve of the specimens in strong fields (Figure 4) shows the presence of two curie points one around 320 °C and the other at 560 °C, though the TRM has always been normal and has a single curie point of 560 °C. Thus, while it has not so far been possible to account for the partial change in the natural moment at 270 °C or to give the reason for the lower 320 °C curie point in high fields, we cannot with certainty say that the magnetic material in these rocks is homogeneous. It is of interest to note that a detailed study of the magnetic interactions in the magnetic minerals of the Linga rocks, undertaken by Lewis*, showed that at least two sets of curie points (one within 200–300 °C and the other around 560 °C) are present and that the partial thermoremanence of the rocks, when the field is applied only above the low curie point range, results in a reversal of magnetic moment at room temperature. It therefore appears that we cannot rule out the possibility of a self-reversing mechanism in the Linga traps at least.

However, there is always the possibility that geological dating of the Deccan traps might not be so conclusive in differentiating between traps separated by vast distances. Perhaps more refined techniques of dating, for example radioactive methods, are essential before one can correlate the palaeomagnetic directions with the time of formation of the traps.

## 6. Acknowledgments

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* Private communication from M. Lewis, Imperial College of Science and Technology, London.

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